

Trainings ▾

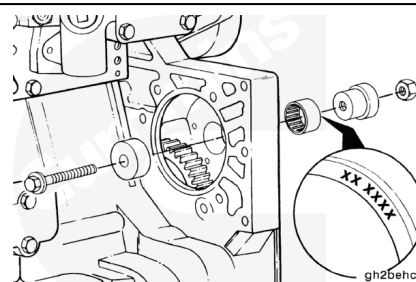
Install

With EGR

If the needle bearing was removed, use bearing installation tool, Part Number 3824117, to install a new needle bearing in the gear housing.

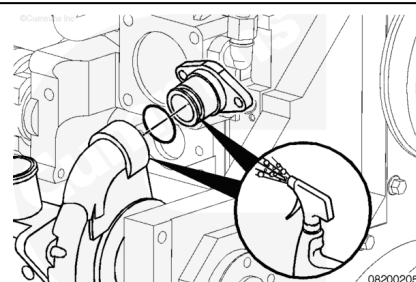
The bearing **must** be installed with the part number side of the bearing against the installation tool to prevent damage to the bearing during the installation.

Install the bearing from the front side of the gear housing until the bearing is flush with the front edge of the housing bore.



⚠ CAUTION ⚠

In order to reduce the possibility of pinching or rolling the o-rings during assembly do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.



Install the water pump cover. Use a new o-ring.

Torque Value: 47 n•m [35 ft-lb]

Install the water pump water transfer connection into the water pump.

Install a new o-ring on the water pump mounting flange.

The water pump **must** be twisted outward from the top until the transfer outlet clears the thermostat housing support during installation.

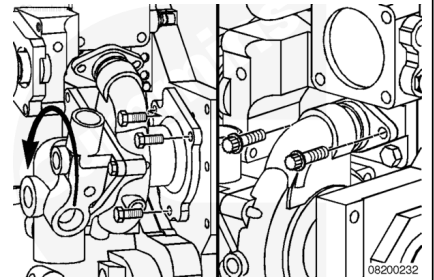
Install the water pump.

Twist the water pump inward and install the three water pump mounting capscrews.

Torque Value: 47 n•m [35 ft-lb]

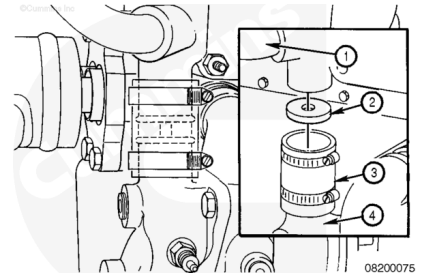
Install a new gasket on the water pump water transfer connection. Install and tighten the water transfer connection capscrews.

Torque Value: 25 n•m [221 in-lb]



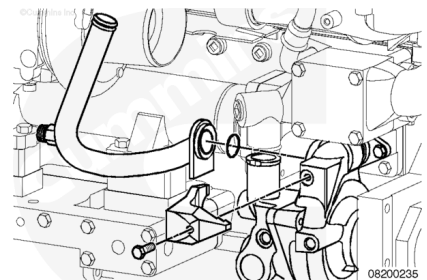
If the engine is equipped with a torque converter cooler, install the disc in the bypass hose before installing the thermostat housing.

1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.



⚠ CAUTION ⚠

In order to reduce the possibility of pinching or rolling the o-rings during assembly do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.



Install a new o-ring on the EGR cooler coolant supply lower tube.

Insert the EGR cooler coolant supply lower tube into the water pump body.

Make sure the EGR cooler coolant supply lower tube is seated completely in the water pump body before tightening the EGR cooler coolant supply lower tube retaining clamp capscrew, to reduce the possibility of pinching the o-ring.

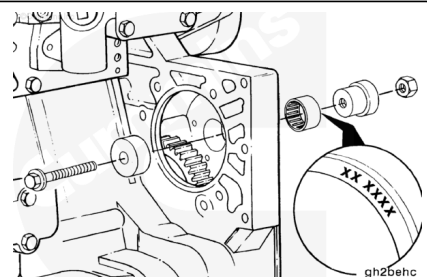
Tighten the capscrew.

Torque Value: 34 n•m [25 ft-lb]

Without EGR

If the needle bearing was removed, use bearing installation tool, Part Number 3824117, to install a new needle bearing in the gear housing.

The bearing **must** be installed with the part number side of the bearing against the installation tool to prevent damage to the bearing during the installation.



Install the bearing from the front side of the gear housing until the bearing is flush with the front edge of the housing bore.

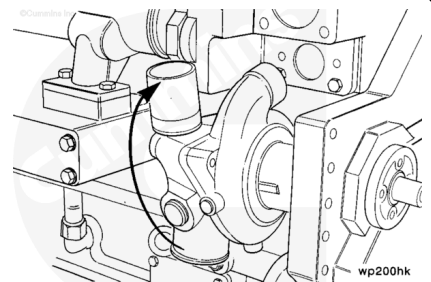
Install the water pump cover. Use a new o-ring.

Torque Value: 47 n•m [35 ft-lb]

Install a new o-ring on the water pump mounting flange.

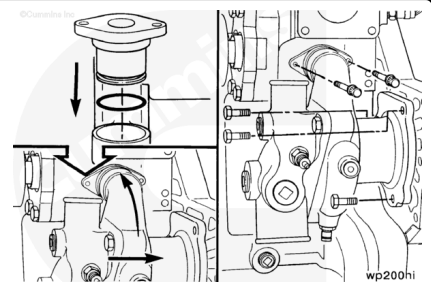
The water pump **must** be twisted outward from the top until the transfer outlet clears the thermostat housing support during installation.

Install the water pump.



⚠ CAUTION ⚠

In order to reduce the possibility of pinching or rolling the o-rings during assembly do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.



Install a new o-ring on the water transfer tube.

Install the connection into the water pump.

Twist the water pump inward and install the three water pump mounting capscrews.

Torque Value: 47 n•m [35 ft-lb]

Install a new gasket on the water pump water transfer connection. Install and tighten the water transfer connection capscrews.

Torque Value: 25 n•m [221 in-lb]

If the engine is equipped with a torque converter cooler, install the disc in the bypass hose before installing the thermostat housing.

1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.

